Text-Guided Diverse Image Synthesis for Long-Tailed Remote Sensing Object Classification

Haojun Tang, Wenda Zhao, *Member, IEEE*, Guang Hu, Yi Xiao, Yunlong Li, and Haipeng Wang

***Abstract*— Remote sensing datasets pose long-tailed data distribution, and such unbalanced datasets will reduce the performance of existing remote sensing object classification models. Existing methods mainly rely on resampling datasets, modifying loss functions, data augmentation, and transfer learning to cope with such challenges. Unlike these, our study takes a novel perspective and focuses on mitigating the long-tailed distribution problem by generating a large number of tail-class images with consistency and diversity. Specifically, this article introduces a novel text-guided tail-class generation network (TGN). TGN comprises two main components: knowledge mutual distillation network (KMDN) and class-consistent diverse tail-class generation network (CDTG). KMDN resolves the isolation issue of the head and tail knowledge by facilitating mutual learning of feature representations between the head and tail data, thereby improving the feature extraction capability of the tail model. CDTG focuses on generating class-consistency diverse tail-class images that uses tail-class features extracted by KMDN. Especially, the class consistency is guaranteed by contrastive language–image pre-trainings (CLIP's) powerful text–image alignment capability. These generated images are then added back into the original dataset to alleviate the long-tailed distribution, thereby improving the tail-class accuracy. Extensive experiments on the widely used DIOR, FGSC-23 and DOTA datasets demonstrate that the proposed method outperforms state-of-the-art methods. Dataset and code are publicly available at https://github.com/XinR-Tang/TGN.**

***Index Terms*— Class-consistent diverse image synthesis, knowledge mutual distillation, long-tailed remote sensing object classification.**

I. Introduction

DEEP neural networks have achieved remarkable success in remote sensing object classification [1], [2], [3], [4], [5]. However, existing datasets, such as DIOR [6], FGSC-23 [7], DOTA [8] typically exhibit long-tailed characteristic, where the number of head-class images is far greater than that of tail-class images. Such unbalance datasets will reduce the performance of most existing remote sensing models. Therefore, how to alleviate this imbalance to ensure the performance of the model for tail classes is a worthy research.

Currently, the academic community primarily addresses the long-tailed object classification from four aspects: resampling [9], [10], reweighting [11], [12], data augmentation [13], [14], and transfer learning [3], [15]. These methods improve the performance on the tail class, thereby improving the overall accuracy. However, they also have some drawbacks. Resampling methods may be divided into under-sampling of the head data [16] or over-sampling of the tail data [10]. The under-sampling degrades the performance of head-class data. Over-sampling usually samples the tail class repeatedly. When the model is trained on these repeated samples, it may learn features specific to these repeated samples rather than learning patterns that generalize to new data. This can lead to overfitting of the tail data. Reweighting methods improve the performance by modifying the loss function that forces the model to pay attention to the tail classes, but it sacrifices the good feature representations of the head classes [11]. Data enhancement is a simple and practical technique for most long-tailed classification tasks. However, some work [17], [18] has shown that simple class-agnostic data enhancement is disadvantageous for long-tailed problems, because such techniques use the same enhancement strategy to process all data, regardless of its category. As images are added for both the head classes and the tail classes, the model still tends to overlearn the knowledge of head classes and neglect the learning of tail classes, which increases the risk of overfitting the tail classes. In recent years, many methods improve the long-tailed problem by using transfer learning, which usually seek some referable knowledge in the source domain to enhance the performance of the tail classes [3], [15].

Resampling is simple and uncomplicated. Reweighting allows the network to focus on the tail classes by itself with a predesigned loss function, without providing additional operations. Many class-aware data augmentation methods [19], [20] also solve the shortcomings of class-agnostic data augmentation and show powerful long-tailed processing capabilities. Transfer learning can learn more from other knowledge to assist long-tailed classification. However, according to our observation, most of the existing work is limited to a pre-determined way, that is, to design models that

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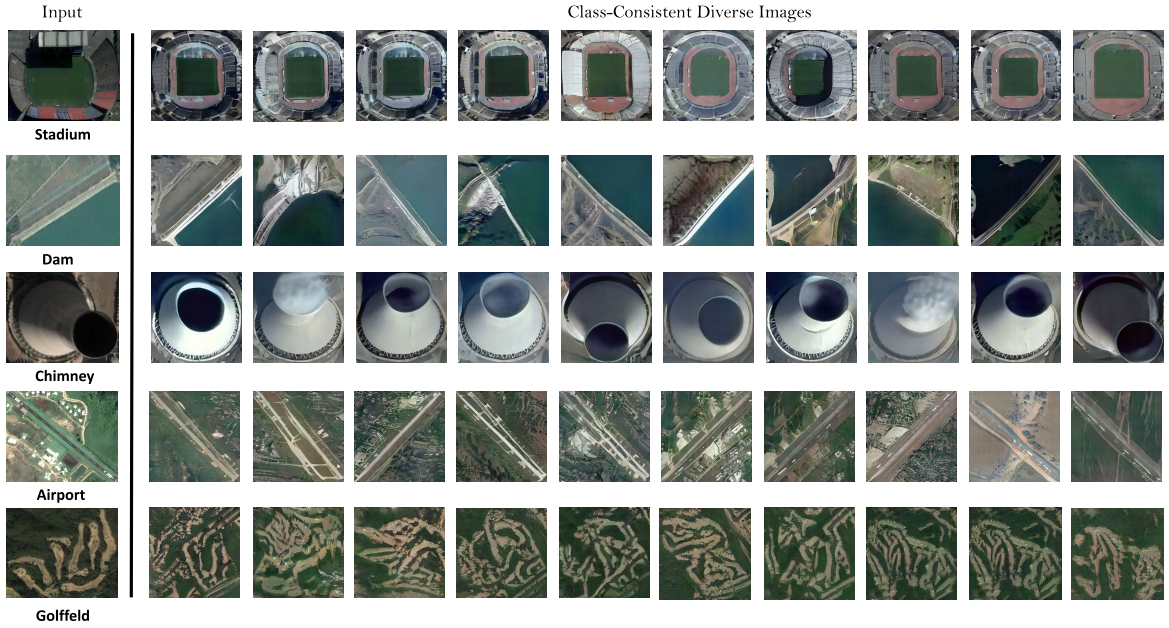


Fig. 1. Different tail-class images generated by the proposed method on the DIOR dataset. Class-consistent diverse images are well generated.

rely entirely on visual modality, which completely ignores the semantic features of the original label text. In addition, most remote sensing classification models are based on the original dataset, and there is no broader sample to be introduced to assist. Therefore, if we can generate a large number of remote sensing satellite images with class consistency and class diversity to alleviate the imbalance of the original dataset, the classification accuracy of long-tailed datasets will be improved. In recent years, a large number of remote sensing image generation methods [21], [22], [23] have been proposed. However, due to the limited number of tail classes, the existing methods are unable to generate class-consistency and class-diversity tail-class images well. Under the circumstances, the semantic features of the label can help.

In this article, we propose a new text-guided tail-class generative network (TGN) to generate scarce class-consistency and class-diversity tail images, thereby improving long-tailed remote sensing object classification. TGN is divided into knowledge mutual distillation network (KMDN) and class-consistent diverse tail-class generation network (CDTG). KMDN aims to mutually learn feature representations from head data and tail data, thereby improving the feature extraction capability of the tail model. CDTG focuses on using the extracted tail features to generate class-consistent diverse tail-class images. Especially, the class consistency is guaranteed by contrastive language–image pre-trainings (CLIP's) powerful text–image alignment capability [24]. Fig. 1 shows the tail-class samples generated by our method. Class-consistent diverse images are well generated.

Specifically, we first divide the long-tailed dataset into two classes (head class D_H and tail class D_T). Then, we design two models, the head model M_H and the tail model M_T, and perform step-by-step training using D_H and D_T, respectively. M_H can learn good feature representations of the head class, but does not have the ability to extract tail features. On the contrary, due to the lack of tail-class data, the feature extraction ability of M_T is relatively poor. In order to address their shortcomings, KMDN first makes M_T pass tail-class feature representations to M_H, and then in turn uses M_H to improve the feature extraction ability of M_T for tail features. After extracting the tail features, we send the features and noise to CDTG. CDTG generates class-consistent diverse tail-class images by adopting CLIP's powerful text–image alignment capability. Finally, these images are used to augment the original dataset to mitigate the long-tailed distribution, eventually improving the long-tailed remote sensing object classification. Fig. 2 shows the distribution of the original dataset and augmented dataset. The accuracy is improved 4.12% after we introduce the generated samples.

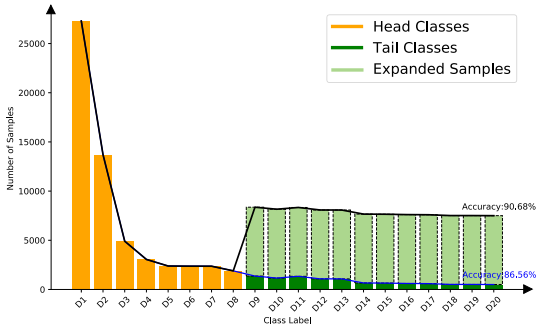


Fig. 2. Data distribution on the original DIOR dataset and adding our generated tail-class images. Accordingly, the accuracy of the classifier improves 4.12%.

Main contributions of this article are summarized as follows.

1) KMDN is proposed to solve the isolation problem of head-class and tail-class knowledge. The tail-class knowledge is distilled to the head-class model that learns the feature representations of the tail classes. Then, the good feature extraction ability of the head-class model is distilled to the tail-class model.
2) CDTG is designed to generate class-consistency diverse tail images, thereby expanding the number of tail-class

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images effectively. CLIP's powerful text–image alignment capability is adopted to guarantee the class consistency.

3) A large number of experiments demonstrate the excellent performance of our approach that mitigates long-tailed remote sensing object classification.

II. Related Work

A. Long-Tailed Object Classification

Due to the extreme imbalance of dataset distribution [25], [26], models trained on such datasets always perform poorly on tail classes. The current methods solve the problem of long-tailed object classification mainly include resampling [27], [28] and reweighting [29], [30]. They all force the model to focus on the tail classes, thus improving the overall performance. Resampling methods generally address this problem from two perspectives: undersampling the head classes and oversampling the tail classes. Researchers have proposed various sampling methods for the original data from different perspectives. Xu et al. [27] introduce a bilateral sampling network. This model mitigates the impact of long-tailed data by considering the semantic relations of coarse and fine granularity. It first performs uniform and inverse sampling on the original data, and then extracts sample features through a unified network. Subsequently, it designs a knowledge transfer strategy from coarse to fine granularity, enabling the model to learn more features of the tail classes. Zhang et al. [28] propose a gradient harmonizing mechanism combined with class adaptability precision, which aims at addressing the imbalance between classification difficulty and sample size in long-tailed problems. This method optimizes the decision boundary through intra-class balance and corrects decision boundary shifts through inter-class balance. However, the current resampling methods are only applied to the original data without introducing new samples, which limits their effectiveness in addressing the long-tailed issue.

Reweighting methods typically involve modifying the loss function to make the network focus more on tail classes. Li et al. [29] propose a feature balance loss, which improves the generalizability by increasing the intensity of feature stimulation for tail classes. Zhang et al. [30] propose a hierarchical re-balancing dual classifier model, which employs a dual classifier framework and uniform sampler, as well as a hierarchical re-balancing loss function, to improve the accuracy of tail-class classification. Although existing reweighting methods have improved classification of the tail classes, they may sacrifice the good performance on the head classes.

B. Transfer Learning

Transfer learning is an effective technique for addressing the long-tailed problem. Generally, it leverages good feature representations from the source domain to guide learning in the target domain. For example, Zhao et al. [3] propose multilevel teacher joint training that uses the well-represented features of head and middle teachers to guide the learning of tail teachers. Chen et al. [31] introduce an integrated network that uses building outlines as auxiliary information. With the designed soft labels, the network is robust to noise in building outlines, thereby mitigating the issue of long-tailed distribution. Li et al. [32] introduce a self-distillation framework for long-tailed recognition that can automatically mine label relationships. Additionally, they propose a new distillation label generation module guided by self-supervision. Distillation labels integrate label and data domain information to effectively model long-tail distributions. Ju et al. [33] classify data into multiple subclasses based on prior knowledge, allowing the model to focus more on specific subclass information. Zhao et al. [34] propose a center-wise feature consistency learning mechanism that mitigates the impact of long-tailed distributions by separating head and tail-class feature learning, thereby enhancing recognition accuracy for tail classes. Transfer learning leverages auxiliary knowledge to enhance model performance, holding great research value. However, acquiring additional auxiliary knowledge requires more effort, especially in the field of remote sensing satellites.

Traditional approaches use the abundant information from head classes to guide the learning of tail classes. However, this one-way distillation method overlooks the implicit knowledge in tail-class images. If the knowledge from tail classes is appropriately transferred to the head model, the head model can learn and subsequently emphasize these implicit representations during knowledge distillation, thereby enhancing the feature extraction ability of the tail model. Therefore, we propose a KMDN, where the tail model first imparts implicit knowledge to the head model. After the head model learns and subsequently strengthens these implicit representations, it guides the tail model to enhance its feature extraction capabilities.

C. Data Augmentation

Data augmentation is a common method for dealing with imbalanced datasets. Zhao et al. [35] propose a style-content metric learning framework to enhance generalizable remote sensing object recognition by focusing on content-based decisions rather than style variations. Li et al. [36] introduce a meta-semantic augmentation (MetaSAug) for enhancing tail-class data. MetaSAug estimates the covariance matrix of tail classes through meta-learning, overcoming the difficulty of estimation due to the scarcity of tail class samples. This enables MetaSAug to generate more accurate and diverse samples for tail classes. Yang et al. [37] propose a location-based scene reconstruction data augmentation method for long-tailed recognition. This method uses the gradient localization technique to increase the scenarios of tail-class samples, enhancing the model's ability to distinguish between head and tail categories, thereby improving the accuracy of long-tailed classification. However, simple class-agnostic data enhancement is disadvantageous for long-tailed problems, which may cause overfitting of the head class.

In recent years, some scholars research the generation of remote sensing satellite images by generative adversarial networks (GANs) [38]. Yates et al. [21] discuss the generation of synthetic aerial imagery and its ability to deceive human observers. By training state-of-the-art unconditional GAN models and using metrics like Fréchet inception distance

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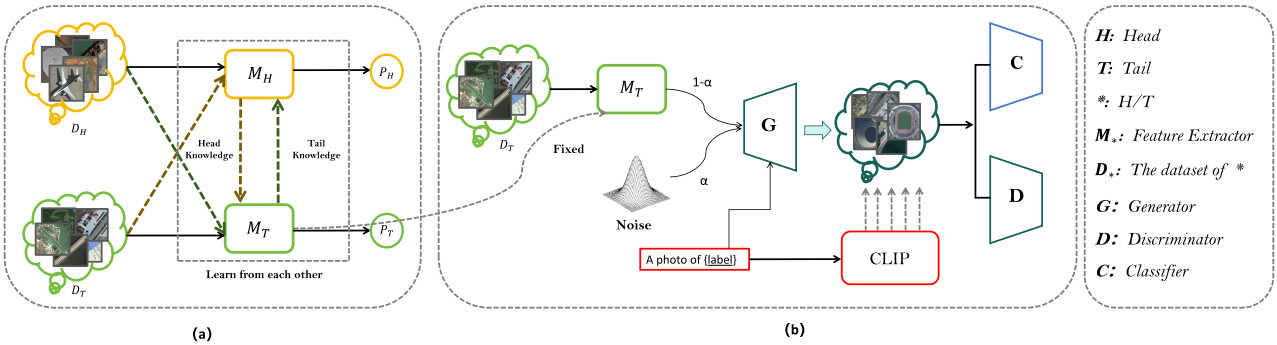


Fig. 3. Architecture diagram of the proposed method. (a) KMDN resolves the isolation issue of the head and tail knowledge by facilitating mutual learning of feature representations between the head and tail data. (b) CDTG uses the features of tail classes extracted by KMDN to generate class-consistent diverse tail-class images.

(FID) and kernel inception distance (KID) for quantitative assessment, they find that these synthetic images could effectively trick human perception. Kourmouli et al. [23] explain the latent space of satellite image GANs that preserves locality. This approach captures the large-scale spatial and spectral variability in satellite images, and decomposes the weight space of pre-trained GANs by maintaining locality. It recovers interpretable directions corresponding to high-level semantic concepts, such as urban structural density or vegetation presence, to guide the synthesis of satellite images. When compared with traditional principal component analysis methods, this approach generates vectors that are better at preserving category information.

However, existing methods often fail to ensure class consistency and diversity effectively. This is primarily because most remotely sensing pre-trained GANs are unsupervised and incapable of generating specific-category remote sensing images. Additionally, the capacity to extract specific features to certain categories is not fully investigated, making it challenging to control the generation of specific tail-class images. These limitations make existing methods be less suitable for addressing the long-tailed problem. To resolve this issue, we utilize the textual knowledge of CLIP and a classifier to ensure the class consistency of the generated images. We provide class diversity by blending the features of tail classes extracted from the proposed KMDN with weighted noise. This approach allows for precise control over the generation of tail-class images, ultimately balancing the long-tailed datasets and training a more accurate classifier. By integrating textual understanding with feature extraction and generative processes, our method effectively addresses the challenges of specific tail-class classification.

III. Proposed Method

In this section, we will begin with a general overview of our motivation and framework, followed by a detailed description of the KMDN and the CDTG, and finally the optimization strategy.

A. Motivation and Framework

Existing schemes cannot alleviate the imbalance of the dataset in terms of expanding the specific tail-class images. However, the semantic knowledge of the text can supervise the process of generation and control the semantic consistency of the image. Different textual descriptions also help to distinguish visually similar objects, so as to better control the generation of tail-class images. So, we propose a text-guided method to generate semantically consistent images through the supervision of text.

To sum up, we propose a novel text-guided tail-class generation network (TGN). It consists of two parts: KMDN and CDTG. KMDN aims to fully extract tail-class features, while CDTG utilizes the CLIP's powerful textual knowledge to generate class-consistent diverse images. Fig. 3 illustrates the overall framework of our method.

1) Given a head model M_H and a tail model M_T, we use the head data $D_H = \{(x_1^H, y_1^H), (x_2^H, y_2^H), \ldots, (x_n^H, y_n^H)\}$ (n represents the total number of samples in the head class) and tail data $D_T = \{(x_1^T, y_1^T), (x_2^T, y_2^T), \ldots, (x_m^T, y_m^T)\}$ (m represents the total number of samples in the tail class) to train them, respectively. Then, M_H and M_T learn their respective feature representations from each other. Specifically, the implicit feature representation of the tail class will be learned by M_H, and then M_H guides the tail model M_T to extract more effective features of the tail class.
2) After the tail-class features are extracted, Gaussian noise is added to provide class diversity. Then, we use the textual knowledge of CLIP to semantically constrain the generated images while letting the classifier discriminate the classes of the generated images, thereby ensuring class consistency. Finally, the discriminator is used to constrain the distribution of generated images.
3) When the overall framework is trained, we only need to give one image and add different scale noises to generate different samples. These samples are class-consistent and diverse, thus expanding the tail-class samples to mitigate the long-tailed characteristic.

B. Knowledge Mutual Distillation Network

Due to the inherent nature of long-tailed data distribution, the head category has a larger sample number, and conversely, the tail category has a smaller sample number. This difference results in better feature extraction ability for M_H and poorer feature extraction ability for M_T. Some existing works [3], [32], [39] have shown that the rich knowledge of head classes

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has a helpful effect on the feature extraction ability of tail classes. However, this unidirectional distillation fails to fully leverage the implicit knowledge in the tail data. M_H has not learned the feature representation of the tail class, so the knowledge of the head class may not be well adapted to the tail-class features and the effect may be limited for M_T. To solve this problem, we first need to let the well-trained head model learn the feature representation of the tail class. Specifically, it is necessary to freeze the pre-trained M_T and then transfer its knowledge to M_H. The pre-trained M_T is subsequently fine-tuned by M_H to learn feature representations suitable for tail-class images.

Therefore, we propose KMDN. The distillation process can be divided into three stages. In the first stage, M_H and M_T learn their own feature representations on the corresponding datasets (D_H and D_T). In the second stage, we need to let M_H learn the feature representation of the tail class. M_T will be frozen and then transfer the knowledge to M_H. In the last stage, M_H teaches the learned tail-class feature representation to M_T. Throughout the process, M_H and M_T rotate the roles of teachers and students. They learn from each other and eventually improve the feature extraction ability of M_T.

As shown in Fig. 3(a), we first train two networks separately. M_H learns the feature representation of D_H, and M_T learns the feature representation of D_T. They have the same network architecture. Next, the extracted features are fed into fully connected layer to obtain the output probabilities P_H and P_T. The cross-entropy loss L_H and L_T used for training M_H and M_T are as follows:

$$L_H = -\frac{1}{N_H} \sum_{x_H \in D_H} y_H \log(P_H(x_H)) \tag{1}$$

$$L_T = -\frac{1}{N_T} \sum_{x_T \in D_T} y_T \log(P_T(x_T)) \tag{2}$$

where N_H represents the total number of samples in D_H, and N_T represents the total number of samples in D_T. x_H and x_T stand for a single sample in the corresponding dataset. y_H and y_T represent binary indicators, used to denote whether the sample belongs to a specific head or tail class. If the sample x_H is a member of the specified head category, $y_H = 1$; otherwise, $y_H = 0$. Similarly, if the sample x_T is a member of the specified tail category, $y_T = 1$; otherwise, $y_T = 0$. L_H and L_T quantify the divergence between the actual and predicted distributions. The values of P_H and P_T range from 0 to 1, leading to invariably negative logarithmic function values. Consequently, when the overall loss is computed by taking the negative sum, L_H and L_T remain nonnegative, ensuring that the loss metrics are meaningful.

After training the two networks separately, M_H has already learned the good feature representations of D_H, and M_T has also learned the corresponding feature representations of D_T. In order to extract more effective features of the tail classes, we need to let M_H learn the tail feature representation. We freeze the parameters of M_T and fine-tune M_H. In this process, M_H will learn implicit knowledge and use its rich head-class knowledge to enhance tail feature representation. Specifically, we extract the features of the corresponding layer in the network and let them close the gap in the corresponding feature space, and the total loss function is represented as follows:

$$L_{M_H} = L_H + L_{\text{dis1}} \tag{3}$$

where L_{M_H} represents the total loss function of M_H, L_H is the corresponding cross-entropy loss of the previous stage, and L_{dis1} is the distillation loss for M_H to learn feature representation extracted by M_T, which can be defined as follows:

$$L_{\text{dis1}} = \frac{1}{N_H} \sum_{x_H \in D_H} (F_H(x_H) - F_T(x_H))^2 \tag{4}$$

where F is the feature representation extracted by the corresponding model from sample x_H. In the distillation process, M_T is frozen and M_H is fine-tuned by M_T. The updates to M_H are constrained not only by the distillation loss L_{dis}, but also by the classification loss L_H. L_H imposes strong constraints on the distillation process and suppresses the drift toward tail features.

Finally, we fix the parameters of M_H and then fine-tune the pre-trained M_T. Since M_H has previously learned the tail-class feature representation, M_T will learn more tail-class feature representation from M_H in the distillation process, and this improves the feature extraction ability of M_T. The total loss function of M_T is defined as follows:

$$L_{M_T} = L_T + L_{\text{dis2}} \tag{5}$$

where L_{M_T} stands for the total loss function of tail model, L_T represents the corresponding cross-entropy loss of the previous stage, and L_{dis2} is the distillation loss for M_T to learn feature representation extracted by M_H, which can be defined as follows:

$$L_{\text{dis2}} = \frac{1}{N_T} \sum_{x_T \in D_T} (F_H(x_T) - F_T(x_T))^2 \tag{6}$$

where F is the corresponding feature of the sample x_T. The updates to M_T are also constrained by the classification loss L_T, so the drift toward head features is suppressed. Up to this point, we have strengthened the ability of M_T to extract tail features by the above mutual learning.

C. Class-Consistent Diverse Tail-Class Generation Network

Now, we can use the extracted tail-class features F_T to generate class-consistent and class-diverse tail-class images. Specifically, we provide diversity by adding Gaussian noise with different weights to the extracted tail-class features, and class consistency is guaranteed by CLIP's powerful text–image alignment capability and a classifier. The detailed architecture is presented in Fig. 4.

First, we use M_T to perform feature extraction on D_T, and then add Gaussian noise to the extracted features, the noise addition process is formulated as

$$\tilde{F}_T = \alpha Z + (1 - \alpha) F_T \tag{7}$$

where $\tilde{F}_T$ represents the tail-class feature after adding noise, Z is the noise obeying Gaussian distribution, F_T is the

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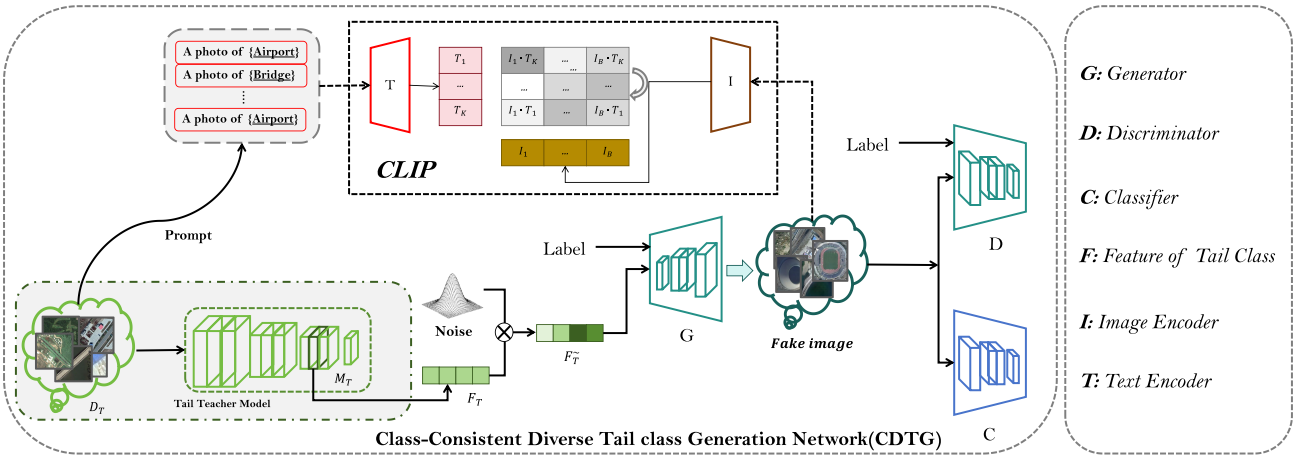


Fig. 4. Framework of the CDTG. The trained M_T is used to extract the tail coarse features, and Gaussian noise is added to provide diversity. Then, the features are fed into the conditional GAN model to generate fake images. The semantic consistency is constrained by CLIP, and the category of the generated image is identified by the classifier. Both of them ensure the class consistency of the generated image.

tail-class feature extracted from M_T, and α is a hyperparameter. By adding different proportions of noise to F_T, we are able to generate diverse samples.

Then, we feed $\tilde{F}_T$ along with the corresponding image's label $y_j^T \in (1, m)$ into the generator to generate fake images

$$D_G = G\left(\tilde{F}_T | y_j^T\right) = \{G_1, G_2, \ldots, G_B\} \tag{8}$$

where G denotes the generator, G_i represents the generated image, B is the size of a batch, and the set of the generated image is denoted as D_G. Since the generator only learns the distribution of the original data and imposes poor constraint on the classes, class confusion may occur. In order to solve this problem, we introduce the knowledge of the text and the classifier to jointly constrain the class consistency of the generated images.

Recently, a large number of textual tasks are used to assist the study of visual tasks. One representative work is CLIP that can evaluate how well images and text match through large-scale contrast learning. This capability allows CLIP to guide the image generation model to create images that match the text description, thus ensuring that the generated images are semantically consistent. In our task, we first pass the generated images through CLIP's image encoder to get image embeddings $I(D_G) = \{I_1, I_2, \ldots, I_B\} \in R^{B\times m}$. Then, we customize a prompt for all the images, e.g., "A photo of Label," where Label is the real label of the corresponding image y_j^T. After that, the prompt is passed through CLIP's text encoder to get the text embeddings $T = \{T_1, T_2, \ldots, T_K\} \in R^{K\times m}$. Here, B represents the number of a batch, K represents the number of prompts, and m is the dimension of an image or text feature. Therefore, we can measure the similarity between an image and a text using the following equation:

$$\text{sim}(I_i, \tilde{T}_j) = \frac{I_i \cdot \tilde{T}_j}{\|I_i\| \cdot \|\tilde{T}_j\|} \tag{9}$$

where $\tilde{T}_j$ represents the transpose of T_j, $i \in (1, B)$, $j \in (1, K)$. "$\cdot$" stands for inner product operation. sim stands for the similarity between image and text feature vectors, ranging from -1 to 1. This similarity is assessed using cosine similarity, which evaluates the angular distance between two vectors independently of their magnitudes. This metric effectively captures the degree of alignment between the vectors' orientations.

After defining the similarity between text and image, the contrastive loss function is optimized to maximize the similarity between positive sample pairs $\text{sim}(I_i, T_i)$ (i.e., image embedding and its matching text embedding) and minimize the similarity between negative sample pairs $\text{sim}(I_i, T_j)$ (i.e., image embedding and mismatched text embedding). The contrastive loss is defined as follows:

$$L_{\text{co}} = \frac{1}{B}\sum_{i=1}^{B} -\log \frac{\exp(\text{sim}(I_i, T_i))}{\sum_{j=1}^{K} \exp(\text{sim}(I_i, T_j))} \tag{10}$$

where numerator $\exp(\text{sim}(I_i, T_i))$ is the exponential value of the similarity of the positive sample pair, and the denominator $\sum_{j=1}^{K} \exp(\text{sim}(I_i, T_j))$ is the sum of the exponential similarity values of all K texts and the image. L_{co} evaluates the average ratio of exponentially weighted sums of $\text{sim}(I_i, T_i)$ to $\text{sim}(I_i, T_j)$ across a batch size B. This loss is designed to enhance the similarity of matched image–text pairs and reduce the similarity of mismatched ones. During optimization, L_{co} promotes higher similarity scores for images corresponding to their target text descriptions over those paired with unrelated texts. Minimizing this loss, the model learns to map images and texts into a common feature space, where semantically similar images and texts get closer together.

CLIP is used to impose semantic constraints on the generated images to ensure semantic consistency. In order to further strengthen the class consistency of the generated images, we pass the generated images through a pre-trained classifier (tail classifier trained by KMDN) to further ensure its class consistency. The classification loss is still the cross-entropy loss, which is expressed as follows:

$$L_C = -\frac{1}{B}\sum_{x_G \in D_G} y_T \log(C(x_G)) \tag{11}$$

where y_T represents a binary indicator, used to indicate whether the generated sample x_G belongs to tail class. If the generated sample is a member of the tail class, $y_T = 1$; otherwise, $y_T = 0$. L_C assesses how well the generated images align with tail class.

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TABLE I
STATISTICAL DETAILS OF DIOR DATASET

Subset	Head								Tail											
Class	Ship	Vehicle	Tenniscourt	Storagetank	Baseballfield	Harbor	Windmill	Airplane	Bridge	Overpass	Groundtrackfield	Expressway-Service-area	Basketballcourt	Airport	Chimney	Expressway-toll-station	Stadium	Dam	Golfield	Trainstation
Identification	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
Train Number	27290	13713	4906	3053	2384	2369	2365	1892	1363	1336	1158	1080	1076	661	649	610	595	512	511	502
Test Number	35106	26654	7360	23361	3433	3140	2998	8212	2604	1778	1880	1085	2149	666	1032	688	673	537	575	509

At this point, we introduce the loss function formulas for the discriminator and generator, defined as follows:

$$L_d = \mathbb{E}_{x_T \sim p\text{data}(x_T|y_T)}[\log(D(x_T|y_T))] + \mathbb{E}_{x \sim P_G(x|y_T)}[\log(1 - D(x|y_T))] \quad (12)$$

$$L_g = -\mathbb{E}_{x \sim P_G(x|y_T)}[\log(D(x|y_T))] + L_{\text{co}} + L_C \quad (13)$$

where L_d stands for the loss function of the classical conditional GAN discriminator, $x_T \sim p\text{data}(x_T|y_T)$ represents the sample x_T obtained from the probability distribution of the original tail-class images, and $x \sim P_G(x|y_T)$ represents the sample x obtained from the probability distribution of the generated tail-class images. We add L_{co} and L_C to the original generator loss to get the total generator loss L_g, which encourages CDTG to generate tail-class samples with class consistency and class diversity. Finally, we add the generated new samples to the original dataset to train the classifier, which improves the accuracy of remote sensing object classification.

D. Optimizing

1) Architecture: The two networks of KMDN have the same architecture, which is ResNet50 [24]. ResNet50 consists of four residual blocks followed by a fully connected module used for classification. We use the first three residual blocks as feature extractor, and the feature maps extracted from the third residual block as intermediate features used for distillation. The fourth residual block and the fully connected operation serve as the classification head. CDTG applies style-based generative adversarial network version 2 (StyleGANV2) [48] as the backbone, and we add CLIP and classification head to constrain class consistency.

2) Training: The whole training process is divided into three stages. In the first stage, we utilize L_H and L_T to train M_H and M_T, respectively. In the second stage, we fix M_T and use its feature F_T to train M_H, followed by fixing M_H and using its feature F_H to train M_T. In the third stage, we use L_g and L_d to train the entire CDTG network.

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: We evaluate the proposed method on three publicly available long-tailed datasets: DIOR [6], FGSC-23 [7] and DOTA [8]. The DIOR dataset contains 20 categories with 192 465 instances. We use 68 025 images as the training set and 124 440 images as the test set, with the number of each category in the training set ranging from 502 to 27 290 (see Table I for details). The FGSC-23 dataset contains 23 categories, all of which are fine-grained ships, with 3256 images as the training set and the remaining 825 images as the test set. The number of each category in the training set varies from 17 to 434 (see Table II for details). The DOTA dataset contains 127 759 images in 15 categories, with 98 906 images in the training set and 28 853 images in the test set. The minimum number of classes in the training set is 325 and the maximum number of classes is 28 068 (see Table III for details). We sort the three datasets in descending order by the number of training sets and then calculate the accuracy for each class.

2) Implementation Details: We implement the experiments on a PC with one RTX 4090 Ti GPU via PyTorch [24]. KMDN is trained with learning rate set to 0.001 and batch size set to 64. Adam optimizer is selected. All images are cropped to the size of 224 × 224. M_H and M_T are first trained separately for 100 epochs, then M_T is fixed to train M_H for 100 rounds, and vice versa. CDTG is trained with the learning rate of 0.002 and the batch size of 8. α is 0.95. Both the generator and discriminator are optimized using Adam optimizer, and the size of the generated images is 256 × 256. The whole network is iterated 200 000 times on both datasets.

B. Ablation Study

1) Effectiveness of KMDN: In Section III-B, we propose the KMDN algorithm to allow M_H and M_T learn their respective feature representations from each other, which ultimately improves the feature extraction capability of M_T. To validate the effectiveness of KMDN, we compare the Top-1 accuracy of M_H and M_T trained directly on the original dataset with the ones after performing mutual learning, and give the corresponding overall accuracy and F1-scores of the head and tail classes. It is worth noting that we train the two networks separately, i.e., M_H is only trained with D_H, and M_T is trained with D_T, to facilitate our comparison of improvement effect before and after KMDN. The experimental results are shown in Table IV, and we can find that with KMDN, the overall

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TABLE II
STATISTICAL DETAILS OF FGSC-23 DATASET

Subset	Head							Tail															
Class	Destoryer	Non ship	Bulk carrier	Frigate	Cruiser	Submarine	Auxiliary ship	Aircraft carrier	Oil tanker	Amphibious assult ship	Combat boat	Hover craft	Landing craft	Fishing boat	Container ship	Liquefied gas ship	Amphibious transport dock	Command ship	Tarawa class amphibious assult ship	Passenger ship	Car carrier	Bange	Medical ship
Identification	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23
Train Number	434	387	274	236	234	190	180	132	132	123	114	96	86	82	81	74	72	71	70	70	58	43	17
Test Number	108	97	69	59	59	48	45	34	33	31	29	24	22	20	20	20	18	18	18	18	14	11	10

TABLE III
STATISTICAL DETAILS OF DOTA DATASET

Subset	Head						Tail								
Class	Ship	Small-Vehicle	Large-Vehicle	Plane	Harbor	Storage-Tank	Tennis-count	Bridge	Swimming-Pool	Helicopter	Basketball-court	Baseball-diamond	Roundabout	Soccer-ball-field	Ground-track-field
Identification	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15
Train Number	28068	26126	16969	7971	5983	5029	2367	2047	1736	630	515	415	399	326	325
Test Number	8960	5438	4387	2531	2090	2888	760	464	440	73	132	214	179	153	144

TABLE IV
EFFECTIVENESS STUDY OF KMDN USING TOP-1 ACCURACY (%) AND F1-SCORE ON DIOR

	Head								Tail											
Setting	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
Training M_H and M_T with D_H and D_T, respectively.	88.06	95.47	**83.45**	70.13	**91.28**	89.98	80.23	97.88	66.17	95.81	**95.80**	**78.35**	**93.64**	82.66	87.35	80.24	**94.29**	96.70	80.26	89.59
	Overall accuracy of M_H: 88.12								Overall accuracy of M_T: 85.05											
	F1 Score of M_H: 82.45								F1 Score of M_T: 79.01											
Training M_H and M_T using **KDMN**	**90.06**	**96.83**	82.95	**70.97**	91.23	**91.38**	**81.85**	**98.27**	**75.08**	**96.70**	95.32	72.55	93.46	**83.72**	**88.95**	**80.98**	93.24	**97.39**	**82.31**	**89.98**
	Overall accuracy of M_H: **89.66**								Overall accuracy of M_T: **86.28**											
	F1 Score of M_H: **83.98**								F1 Score of M_T: **82.26**											

accuracy of tail classes is improved by 1.23%. The F1-score is improved by 1.53% for the head class and 3.25% for the tail class. The experimental results show that KMDN has the ability to improve the feature extraction ability of M_T.

2) Influence of CDTG: After extracting the tail-class features F_T using M_T, we feed F_T into CDTG to generate class-consistent diverse tail-class images. To verify the effectiveness of CDTG, we first perform image generation on the original StyleGANV2, and then experiment with StyleGANV2 + label, i.e., feed labels of tail classes into the network, and finally experiment with CDTG, i.e., StyleGANV2 adds with the CLIP and the classifier. The images generated from the experiments are shown in Fig. 5, and it can be found that the result of StyleGANV2 shows

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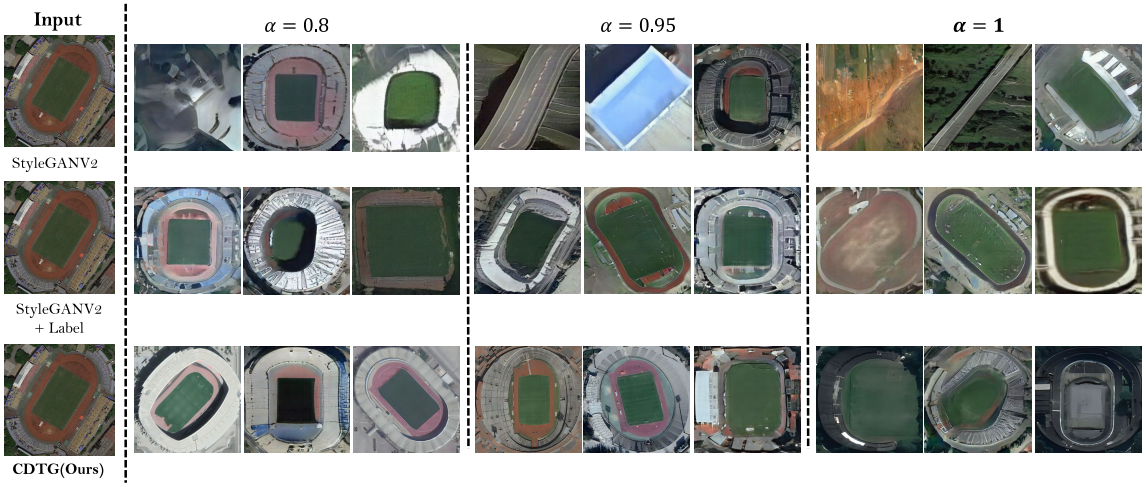


Fig. 5. Generation results of StyleGANV2, StyleGANV2 + label, and CDTG. StyleGANV2 can lead to class confusion. StyleGANV2 + label has some effect on controlling the class, but the generated images of class stadium and class ground track field are similar. In contrast, our method is able to maintain class consistency and generate samples with diverse visual effects by blending different proportions of noise. When there is only noise input ($\alpha = 1$), the semantic features do not play a role, so the generated samples are out of frame. But the generated samples by our method have both class consistency and class diversity.

TABLE V

IMPACT STUDY OF THE EXPANDED SAMPLES USING TOP-1 ACCURACY (%) AND F1-SCORE ON DIOR. AVERAGE REPRESENTS THE RATIO OF THE NUMBER OF CORRECTLY PREDICTED SAMPLES TO THE TOTAL NUMBER OF SAMPLES IN THE CORRESPONDING DATASET. A* IS THE NUMBER OF EXAMPLES ADDED TO THE TAIL CLASS. U* IS THE NUMBER OF EXAMPLES ADDED TO THE HEAD AND TAIL CLASS

			Head								Tail											
Setting	F1	Average	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
A0	79.99	86.56	87.65	89.23	93.22	87.27	86.28	75.41	98.10	72.55	**73.89**	61.92	**95.32**	91.43	92.42	88.74	82.85	**83.14**	72.51	74.12	**96.87**	83.89
A1000	81.73	87.42	83.33	91.44	94.78	**91.79**	87.79	75.89	94.26	**90.90**	58.10	74.13	91.06	92.72	92.79	92.49	80.23	72.24	51.56	59.96	92.70	72.30
A3000	84.75	89.86	89.84	97.02	**95.20**	87.01	91.44	78.98	98.93	79.86	71.24	71.77	95.21	**94.29**	94.88	91.74	**84.69**	78.63	79.20	75.23	96.35	**88.21**
A5000	85.10	90.61	89.27	96.95	93.59	90.10	90.59	**80.57**	**99.10**	85.89	73.16	**73.62**	94.89	93.00	**95.58**	93.39	80.52	75.73	78.90	**75.42**	95.83	88.02
A7000	**85.11**	**90.68**	**90.19**	96.98	94.73	90.28	**93.97**	78.82	98.90	83.99	71.16	72.67	94.31	92.81	94.28	**93.84**	82.95	72.97	**80.68**	74.12	96.70	87.62
U7000	82.83	87.74	89.78	**97.26**	94.74	83.56	89.43	80.19	98.40	72.59	70.54	71.44	92.83	91.52	94.70	92.34	84.40	80.09	78.16	72.19	73.74	86.82

class confusion, where the input classes are inconsistent with the generated classes. StyleGANV2 + label also exhibits class confusion for two similar samples, e.g., stadiums and ground track field. On the contrary, our method not only ensures class consistency, but also generate different visualizations by controlling the fusion ratio of noise and F_T.

In addition, we discuss the effect of hyperparameter α. α is used to adjust the proportion of F_T and noise mixing, which controls the diversity and detailed features of the images. When α is closer to 1, the diversity of the generated images increases, but more details are lost. We select three sets of values: 0.95, 0.8 and 1, respectively, and take three generated samples from each set for comparison. In the last row, as F_T ratio increases (i.e., α gets smaller), the diversity decreases and the generated stadium are similar. So, we propose to choose α to be 0.95, which can maintain both semantic feature and diversity.

3) Impact of the Expanded Samples: To verify the effectiveness of our method for mitigating the classification of long-tailed objects, we expand 1000, 3000, 5000, and 7000 images for each tail class, respectively, and then use them to train the classification network. In addition, to compare the difference between class-agnostic data augmentation and our method, we additionally add a set of experiments named U7000 (add 7000 generated images for all categories). The goal is to discuss overfitting due to class-agnostic data augmentation. The experimental results are shown in Table V, and Fig. 6 shows the trend of accuracy changing with the increase in the number of tail-class samples. It can be seen that with the increase in the number of images in the tail class, the accuracy

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TABLE VI

TOP-1 ACCURACY (%) AND F1-SCORE OF DIFFERENT METHODS ON THE TESTING SET OF DIOR DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED

				Head								Tail										
Method	Backbone	F1 Score	Average	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
SADE [40]	ResNet50	79.47	88.40	80.00	90.50	78.63	85.11	91.91	91.73	92.89	98.70	89.19	**95.63**	71.89	89.44	87.34	89.68	79.07	**98.43**	92.34	70.92	83.36	**95.48**
RIDE [41]	ResNet50	78.34	87.57	82.28	85.49	79.20	83.17	92.37	**92.99**	92.25	98.90	**90.69**	95.07	68.20	85.76	82.12	91.43	72.09	93.04	91.22	66.99	73.25	88.41
ResLT [42]	ResNet50	64.34	72.95	89.98	78.40	52.27	99.00	84.82	89.09	71.24	55.64	77.04	83.06	75.00	90.41	85.89	90.00	85.81	81.81	72.27	74.12	72.17	85.07
TLC [43]	ResNet32	70.77	82.70	68.47	78.68	71.75	83.78	87.81	89.02	84.54	94.13	76.58	88.60	46.70	81.78	73.18	83.41	60.03	94.26	89.51	65.92	72.21	84.48
BKD [44]	ResNet32	63.26	75.71	42.58	84.91	69.14	82.66	73.88	77.44	83.84	88.06	61.41	79.11	42.01	67.73	58.29	55.02	52.62	74.09	80.00	48.59	42.79	75.44
LAL [45]	ResNet32	75.09	84.73	57.14	80.48	74.43	88.63	82.70	88.13	94.25	96.63	81.68	91.34	54.07	76.74	78.03	90.05	65.55	84.52	83.09	74.58	52.15	84.09
BALLAD [46]	ResNet50	75.90	87.84	94.34	81.56	**97.34**	97.32	90.44	77.70	57.58	96.12	46.13	60.51	78.74	81.83	81.03	66.89	63.66	30.04	84.03	45.49	90.63	70.88
EFS [47]	ResNet50	78.44	85.41	**95.14**	69.42	91.38	**99.99**	87.12	89.34	61.28	**99.41**	71.71	69.98	71.12	**94.15**	**95.21**	90.06	**92.24**	74.32	**97.11**	73.15	93.11	92.25
TGN **(ours)**	ResNet32	**86.16**	**90.41**	90.64	96.51	93.68	89.51	90.24	77.58	98.77	82.45	71.27	74.92	93.94	93.27	94.32	90.99	81.01	74.56	76.67	**78.58**	94.78	87.82
TGN **(ours)**	ResNet50	**85.11**	**90.68**	90.19	**96.98**	94.73	90.28	**93.97**	78.82	**98.90**	83.99	71.16	72.67	**94.31**	92.81	94.28	**93.84**	82.95	72.97	80.68	74.12	**96.70**	87.62

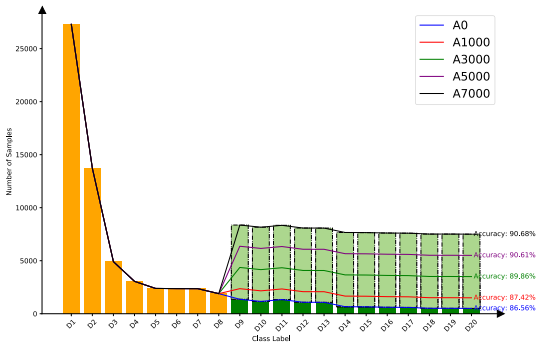


Fig. 6. Different accuracy curves under varying numbers of tail-class samples. The accuracy gradually rises as we add 1000, 3000, 5000, and 7000 tail-class samples.

of the classifier gradually increases and reaches a maximum of 90.68%. The F1-score also gradually increased with the increase of tail samples, and finally reached 85.11%. It is 4.12% higher than that without adding any samples. Finally, we compare the two results A7000 and U7000. We can observe from Table V that accuracy and F1-score both decrease after adding 7000 samples uniformly for all classes, and the decline for the tail classes is more pronounced. Therefore, our method can better alleviate the long-tailed effect.

C. Comparison With the State of the Art

We make a comparison between our proposed method and eight state-of-the-art methods, namely self-supervised aggregation of diverse experts (SADE) [40], routing diverse experts (RIDE) [41], residual learning for long-tailed recognition (ResLT) [42], trustworthy long-tailed classification (TLC) [43], balanced knowledge distillation (BKD) [44], long-tail learning via logit adjustment (LAL) [45], balanced linear adapter (BALLAD) [46], and easy fusion with stacking (EFS) [47]. Among them, the first six methods all belong to visual modality. The last two methods, which are the same as our method and belong to multimodal methods, also introduce the knowledge of text to assist classification. In order to fairly compare the advantages of our method, we first compare TGN with six methods in visual modality, where SADE, RIDE, and ResLT's backbone is ResNet50 and TLC, BKD, and LAL's backbone is ResNet32. We divide our methods into two types: TGN_32 and TGN_50, which are obtained by training on ResNet50 and ResNet32 using the new expanded dataset, respectively. Then, we compare with two multimodal long-tailed classification methods, and here our backbone uniformly choose ResNet50. For all experiments, we use the author's source code and fine-tune on the relevant dataset.

1) Comparison on the DIOR Dataset: The results of different methods on the test set of DIOR dataset are reported in Table VI. We can see that our method has the highest average accuracy of 90.68% and 90.41%, which is 2.01%–17.73% higher than the other methods. Compared with two multimodal methods, our method outperforms BALLAD and EFS by 2.84% and 5.27%, respectively. In particular, the classification of the tail class is improved. Similarly, our method outperforms other methods in the comparison of F1-scores. TGN_32 achieves the highest 86.16%, showing the ability to alleviate the long-tailed characteristics.

2) Comparison on the FGSC-23 Dataset: The results of different methods on the test set of FGSC-23 dataset are shown in Table VII. We can see that our method has the highest average accuracy of 77.70% on ResNet32, which surpasses the suboptimal method by 6.91%. Moreover, our method achieves a better result than BALLAD and EFS which are multimodal methods with the accuracy improvement of 45.39% and 8.77%. Our method outperforms the other

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TABLE VII

TOP-1 ACCURACY (%) AND F1-SCORE OF DIFFERENT METHODS ON THE TESTING SET OF FGSC-23 DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED

				Head							Tail														
Method	Backbone	F1 Score	Average	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23
SADE [40]	ResNet50	58.38	70.79	64.95	67.59	**84.75**	64.41	**81.25**	60.00	57.97	76.47	68.18	66.67	66.67	64.52	61.11	**99.99**	**86.21**	60.00	**92.86**	95.83	66.67	65.00	61.11	95.00	90.91
RIDE [41]	ResNet32	34.91	52.27	73.21	33.33	18.31	28.99	26.76	82.81	35.37	55.56	72.83	35.71	49.41	26.15	39.73	31.71	33.33	60.66	55.88	82.35	89.33	46.97	44.12	94.64	97.06
ResLT [42]	ResNet32	58.81	62.55	62.89	73.15	74.58	61.02	31.88	79.41	**67.74**	83.33	72.41	46.67	27.27	45.45	33.33	61.11	38.89	99.99	35.00	85.71	**99.99**	50.00	50.00	99.99	81.82
TLC [43]	ResNet50	25.24	44.12	46.39	21.30	28.81	23.73	43.75	28.89	10.14	64.71	22.73	61.11	33.33	45.16	61.11	**99.99**	**86.21**	60.00	85.71	91.67	57.58	60.00	66.67	99.99	99.99
BKD [44]	ResNet50	57.82	65.94	57.73	71.30	**84.75**	77.97	34.78	76.47	54.84	72.92	72.41	46.67	27.27	59.09	66.67	61.11	55.56	99.99	70.00	**99.99**	99.99	65.00	61.11	95.00	99.99
LAL [45]	ResNet32	31.29	53.45	75.26	72.22	42.37	18.64	47.83	64.71	58.06	68.75	**79.31**	42.22	21.21	27.27	27.78	27.78	5.56	80.00	50.00	42.86	91.67	40.00	44.44	45.00	99.99
BALLAD [46]	ResNet50	23.79	32.31	38.46	55.17	35.29	33.33	21.28	26.47	11.11	31.57	31.25	29.63	22.41	47.62	20	17.02	18.18	88.88	20.00	2.29	31.25	12.5	8.69	64.29	13.79
EFS [47]	ResNet50	59.82	68.93	73.02	78.35	69.57	**88.14**	74.58	**91.67**	60.00	85.29	63.64	**77.42**	72.41	**100**	72.73	65.00	50.00	**100**	38.39	55.56	83.33	77.78	85.71	90.91	**100**
TGN **(ours)**	ResNet32	**74.32**	**77.70**	**87.04**	**95.47**	71.64	77.08	**87.80**	75.86	46.81	**86.21**	73.91	73.53	**82.61**	85.19	**100**	94.12	58.33	**100**	24.29	24.24	55.00	**87.50**	**100**	**100**	**100**
TGN **(ours)**	ResNet50	**69.62**	**71.76**	83.33	70.10	55.07	84.75	79.66	75.00	57.78	82.35	48.48	64.52	79.31	87.50	59.09	45.00	50.00	**100**	61.11	66.67	66.67	55.56	85.71	90.91	60.00

TABLE VIII

TOP-1 ACCURACY (%) AND F1-SCORE OF DIFFERENT METHODS ON THE TESTING SET OF DOTA DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED

				Head						Tail								
Method	Backbone	F1 Score	Average	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15
SADE [40]	ResNet50	69.64	93.57	92.85	94.81	91.75	98.1	93.97	94.11	92.36	82.57	95.22	89.04	82.57	92.99	**88.27**	**88.23**	86.8
RIDE [41]	ResNet50	58.62	81.54	81.12	87.29	78.72	90.74	84.19	88.51	90.2	92.66	89.25	86.53	76.51	73.64	69.07	73.29	61.24
ResLT [42]	ResNet50	76.94	94.97	96.17	95.5	89.45	96.04	**96.41**	94.9	61.76	**97.57**	92.18	54.55	91.3	82.35	82.14	83.33	90.48
TLC [43]	ResNet32	79.69	88.24	90.78	90.47	84.13	93.32	89.76	81.02	88.29	76.08	92.5	58.9	72.73	**94.86**	85.47	60.13	81.94
BKD [44]	ResNet32	67.64	84.74	87.04	93.78	82.61	80.4	84.31	82.83	81.71	65.09	88.41	36.99	40.15	87.38	15.64	54.9	33.33
LAL [45]	ResNet32	78.31	92.77	96.83	94.1	92.04	96.52	91.53	90.58	84.74	84.05	83.86	45.21	50.76	83.18	51.4	55.56	74.31
BALLAD [46]	ResNet50	62.01	90.80	93.73	95.82	86.99	92.05	88.27	**97.81**	33.33	70.49	79.31	37.03	42.22	27.16	27.47	32.84	27.03
EFS [47]	ResNet50	87.45	95.47	97.97	94.69	**92.54**	**99.13**	95.47	95.98	70.59	78.95	**97.38**	92.98	75.18	87.44	74.52	77.78	81.17
TGN_32(Ours)	ResNet32	**89.75**	**95.56**	**98.32**	97.87	85.42	96.04	94.61	94.94	79.41	93.52	81.43	88.18	69.57	44.12	73.21	61.90	80.95
TGN_50(Ours)	ResNet50	**90.44**	**96.1**	98.14	**98.22**	87.05	97.65	95.41	96.92	**98.63**	95.55	81.11	**92.98**	**82.61**	41.18	72.32	78.57	**90.48**

comparison methods by 14%–50% in F1-score. In particular, we outperform the multimodal method BALLAD by 50.53%. Our scheme is effective for long-tailed data balancing.

3) Comparison on the DOTA Dataset: The results of different methods on the test set of DOTA dataset are shown in Table VIII. The accuracy of our method has increased compared with other comparison methods, especially the classification accuracy of tail class. Meanwhile, our method achieves higher accuracy than BALLAD and EFS which are also combined with the text modal. In terms of F1-score, our method achieves 90.44%, which shows the best performance. Based on all the experiments, we find that the introduction of text is helpful to improve the long-tailed classification accuracy.

4) Discussion: We mitigate the long-tailed characteristic of the dataset by adding a large number of class-consistent diverse tail-class images, and abundant experiments have proved the validity of our method. Second, in the case of categories with very few instances, our method can alleviate

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Fig. 7. Generation results for tail classes on the FGSC-23 dataset. *N* stands for the number of tail samples. The first row shows the input images, while the second to the sixth rows display the generated images.

the issue of mode collapse due to the textual guidance. However, for categories with extremely small sample sizes, mode collapse may still occur. Fig. 7 shows the generation results for tail classes on the FGSC-23 dataset. Even with only 43–86 samples for the tail classes, our method can still generate ships with different poses and colors. However, for classes with only 17 samples, mode collapse still occurred. This is an issue we plan to address in future research.

V. Conclusion

We propose a new method from the perspective of expanding the tail-class dataset. In particular, we propose TGN, which first learns the feature representations of head data and tail data from each other via KMDN, thereby enhancing the feature extraction capability of the tail model. Then, the extracted tail-class features are used to generate a large number of class-consistent diverse tail-class images by CDTG, where the CLIP's text–image alignment capability helps guarantee the class consistency. Experimental results show that our method can effectively alleviate the long-tailed characteristic in aerial images, which is superior to the existing state-of-the-art methods.

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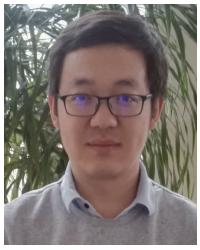
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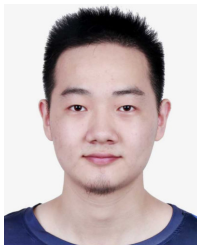
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